

Savannah Church

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EDUCATION

Virginia Tech — Blacksburg, VA | *B.S. in Physics; Minors in Astronomy and Mathematics* Aug 2022 – May 2026
GPA: 3.17

Coursework: Optics, Cosmology, Physics of Galaxies, Thermal Physics, Quantum Mechanics, Astrophysics, Mechanics, Electricity & Magnetism, Chemistry

Labs: Chemical, Electrical, Modern Physics, Observational Astrophysics

RESEARCH & TECHNICAL EXPERIENCE

Researcher | *PolarNOx Mission, Virginia Tech* May 2024 – May 2026

- Designed and machined an aperture casing for an in-lab spectrograph using Altium and CAD in the aerospace machine shop, enabling stable, light-tight test conditions.
- Ran weekly spectrograph tests in a controlled darkroom environment and analyzed the resulting data to calibrate instrument performance ahead of launch.
- Built a Python-based scheduling tool to identify optimal launch date and time windows, supporting mission planning for the sounding rocket.
- Collaborated with a cross-functional team to design, build, and successfully launch a sounding rocket (Jan 30, 2026) that measured nitric oxide levels during polar night using UV stellar occultation.
- Co-authoring publication: "Sounding Rocket Mission Design for Polar Night Nitric Oxide Measurement Using UV Stellar Occultation" (Publication Date TBD).
- Presented mission design at Design Review, Wallops Flight Facility; poster session at the 2025 CEDAR/GEM Conference; and a technical briefing to a graduate industry panel.
- Contributed to optical systems work at LASP (Laboratory for Atmospheric and Space Physics).
- Maintained detailed records of test procedures and design iterations, serving as a reference for teammates during handoffs and manuscript preparation.

Undergraduate Student Researcher | *Dennis Dean Undergraduate Research Conference* Jan 2023 – Apr 2023

- Investigated H1 (atomic hydrogen) emissions related to galaxy formation, analyzing multiple data cubes using Linux-based tools.
- Authored a research paper and presented findings to STEM educators and students at the conference.

ADDITIONAL EXPERIENCE

Cook | *High Country Stables (Rocky Mountain National Park Concessions)* June 2026 – Present

- Prepared 3 meals per day for a stable crew of up to 25 people, managing kitchen organization, grocery procurement, and snack/dessert prep.

Coach | *Skyline Gymnastics* Jan 2021 – Sep 2022

- Instructed children ages 1–10 in foundational gymnastics techniques and guided parents on safe practice methods to support skill development.

LEADERSHIP & SERVICE

- Wesley at Virginia Tech Missions Team (Mar 2025 – May 2026): Organized fundraisers and community service events; tutored elementary/middle school students weekly; participated in hurricane recovery efforts in Augusta, GA and Anderson, SC.
- Secretary, Virginia Tech Club Gymnastics (May 2023 – Jun 2024): Managed communications, scheduling, and logistics for a competitive club team.
- Orion Living/Learning Community (Aug 2022 – May 2023): Developed professional skills and participated in community service.
- VT Science Festival Volunteer Exhibitor (2022): Taught elementary students to build drawing robots; organized and ran an activity station.

SKILLS

Hands-on: Astronomical measurements, bench testing, laboratory instrumentation, optical systems, stellar spectroscopy, vacuum systems, data collection, analytical testing

Technical / Computer: Python, CAD, data analysis, stellar occultation analysis, Linux, Microsoft Office, Claude, Microsoft Copilot, Excel

Other: Basic building & construction, scientific writing, presentation skills

Languages: Elementary Japanese, Elementary Spanish